- A suitable current (1A) is passed through the coil by adjusting rheostat so that the deflection of the aluminium pointer lies between 30° and 60°.
- The value of the current I is noted from ammeter.
- Two readings θ_1 and θ_2 corresponding to two ends of the pointer are noted.
- Now the direction of the current is reversed using commutator, two more readings θ_3 and θ_4 are noted.
- Now the compass box is taken to the other side and is kept at the same distance *x*.
- Four more readings θ_5, θ_6, θ_7 and θ_8 are taken as done before.
- These eight readings and their average value are tabulated.
- The experiment is repeated for another value of current, say 1.5 A by keeping the compass box at the same distance *x*.
- The radius of the circular coil is found by measuring the circumference of the coil using a thread around the coil.
- The number of turns n of the coil is noted.
- From the values of *n, r, x* and *I*/tanθ, the horizontal component of Earth's magnetic field is now found using the formula.

OBSERVATION

Number of turns in the coil *n*=

Circumference of the coil (2πr) =

Radius of the coil r =

To find horizontal component of the Earth's magnetic field

S.No	Distance *x* (cm)	Current I (A)	Deflection for eastern side				Deflection for western side				Mean θ	$\frac{I}{\tan\theta}$
			θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8		
											Mean	

CALCULATION

$$B_H = \frac{\mu_0 n I r^2}{2\left(r^2 + x^2\right)^{3/2}} \left(\frac{1}{\tan\theta}\right) =$$

RESULT

Horizontal component of the Earth's magnetic field at a place = _____T

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